

Project Progress Update #1 Speaker Notes (Slide-by-Slide)

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Project: Predicting Marathon Race Performance Using Machine Learning

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Slide 1 — Title Slide (Sanad)

(Predicting Marathon Race Performance Using Machine Learning)

Today we will present our first project progress update for our IECE 465 machine learning project. Our project focuses on predicting marathon finish times using machine learning techniques.

We are using real-world race data from the 2023 Boston Marathon to build predictive models that estimate a runner's final net finish time. This is formulated as a supervised regression problem where the goal is to predict a continuous variable.

In this presentation we will walk through the dataset, how we processed the data, the machine learning models we implemented, and some preliminary results comparing model performance.

Slide 2 — Agenda (Mahib)

In this presentation we will walk through seven main sections.

First, we will introduce the project motivation and overall objective.

Next, we will describe the dataset and its features.

Then we will explain the preprocessing steps we used to clean and prepare the data for machine learning.

After that we will discuss the two models we implemented: a baseline linear regression model and a neural network regression model.

We will then review preliminary results comparing their performance.

Finally we will demonstrate our system and discuss future improvements and next steps.

Slide 3 — Project Motivation (Brian)

Predicting marathon finish times is an interesting machine learning problem because marathon performance depends on many different factors.

These include demographic factors such as age and gender, physiological factors like endurance, and race strategy factors such as pacing.

In this project, our main objective is to predict a runner's **net marathon finish time** using features available in the race dataset.

To evaluate modeling approaches, we compare two regression models:

First, a **linear regression model**, which serves as our baseline.

Second, a **neural network regression model**, which can capture more complex nonlinear relationships.

Applications of this work include sports analytics, optimizing athlete training strategies, and predicting performance outcomes.

Slide 4 — Dataset Overview (Sanad)

Our dataset comes from the **2023 Boston Marathon results dataset**.

It contains race results for **26,598 runners** and includes **15 variables describing performance and demographic information**.

For this project we selected a subset of the most relevant features.

The key input features we use are:

- Age group
- Gender
- Half-marathon split time

Our target variable is the **net finish time in seconds**, which represents the runner's final race time after crossing the start line.

The dataset also contains additional information such as placement rankings and gun time versus net time, but these were not used directly in our prediction model.

Slide 5 — Dataset Preprocessing (Brian)

Before training our models, we performed several data preprocessing steps.

First, we imported the dataset into MATLAB using the `readtable` function.

Next, we selected the features relevant to our prediction task.

Then we removed rows with missing values using MATLAB's `rmmissing` function to ensure the dataset contained only valid entries.

Finally, we converted categorical variables such as **age group** and **gender** into categorical data types so they could be properly handled by the regression models.

These preprocessing steps ensure the dataset is clean and properly formatted for training machine learning models.

Slide 6 — Data Visualization (Mahib)

Before building models, we explored the dataset through visualization.

Several useful plots can help us understand the data.

For example:

- The distribution of marathon finish times
- The distribution of runners across age groups
- And the relationship between half-marathon split time and final finish time

One of the most important insights we observed is that **half-marathon split time is strongly correlated with the final marathon finish time**.

Because of this strong relationship, a useful visualization is a **scatter plot** where:

- The X-axis represents half marathon time
- The Y-axis represents final finish time.

This relationship forms the basis for our predictive models.

Slide 7 — Model 1: Linear Regression (Sanad)

Our first model is a **linear regression model**, which serves as our baseline approach.

Linear regression assumes a linear relationship between input features and the target variable.

In MATLAB we implemented this model using the `fitlm` function.

The model predicts finish time as a function of **age group, gender, and half-marathon split time**.

After training the model on the training dataset, we used the `predict` function to generate predictions on the test data.

To evaluate performance we used two common regression metrics: **Mean Absolute Error** and **Root Mean Squared Error**.

Slide 8 — Model 2: Neural Network (Brian)

Our second model is a **neural network regression model**.

Unlike linear regression, neural networks can capture **nonlinear relationships between variables**.

Our neural network consists of:

- Input features
- Two hidden layers
- The first hidden layer contains 32 neurons
- The second contains 16 neurons

We used the **ReLU activation function**, which is commonly used in deep learning models.

In MATLAB we implemented this model using the `fitrnet` function.

This model allows us to evaluate whether a more complex nonlinear model improves prediction accuracy.

Slide 9 — Model Evaluation Metrics (Sanad)

To evaluate the performance of our models we used two standard regression metrics.

The first metric is **Mean Absolute Error**, or MAE.

MAE measures the average absolute difference between the predicted value and the true value.

The second metric is **Root Mean Squared Error**, or RMSE.

RMSE penalizes larger errors more heavily because it squares the prediction errors before averaging.

For both metrics, **lower values indicate better predictive accuracy**.

Using both metrics allows us to more accurately compare model performance.

Slide 10 — System Demonstration (Brian)

This slide shows the MATLAB implementation of our system.

The script loads the dataset, performs preprocessing, trains both models, and generates predictions.

It also calculates evaluation metrics and produces visualization plots comparing actual and predicted results.

During the live demonstration we will run this script and show how the models generate predictions from the dataset.

This demonstrates the full machine learning workflow from data loading to prediction.

Slide 11 — Preliminary Results (Sanad)

Here we compare the performance of the two models.

The linear regression model achieved:

- MAE of about **384 seconds**
- RMSE of about **549 seconds**

The neural network achieved slightly lower error values:

- MAE of about **382 seconds**
- RMSE of about **548 seconds**

This means the neural network performs slightly better, although the difference between the models is relatively small.

One important observation is that **half marathon split time dominates prediction accuracy**, since it strongly correlates with the final finish time.

Slide 12 — Actual vs Predicted Results (Brian)

This slide shows scatter plots comparing **actual finish times versus predicted finish times**.

Each point represents one runner.

Points that lie close to the diagonal line indicate accurate predictions.

Both models produce predictions that cluster closely around this line, indicating strong predictive capability.

This visualization confirms that both models capture the relationship between half-marathon splits and final race time effectively.

Slide 13 — Next Steps (Mahib)

There are several improvements we plan to explore moving forward.

First, we will perform **feature engineering**, which may help the model capture additional patterns in the data.

We also plan to incorporate **additional runner characteristics** if available.

Another improvement will be **hyperparameter tuning** to optimize model performance.

We will also implement **cross-validation** to ensure the models generalize well.

In the long term, we aim to improve prediction accuracy, analyze feature importance, and compare additional machine learning models.

Slide 14 — Questions (Sanad)

That concludes our presentation.

We demonstrated how machine learning models can be used to predict marathon finish times using real-world race data.

We showed the full workflow including dataset preparation, model training, evaluation, and preliminary results.

Thank you for your time, and we would be happy to answer any questions.